

Using Explainable AI to Characterize Features in the Mirai Mammographic Breast Cancer Risk Prediction Model

Yao-Kuan Wang, MSc¹ • Zán Klaneczek, PhD² • Tobias Wagner, MSc¹ • Lesley Cockmartin, PhD³ • Nicholas Marshall, PhD^{1,3} • Andrej Studen, PhD^{2,4} • Robert Jeraj, PhD^{5,2,4} • Hilde Bosmans, PhD^{1,3}

Author affiliations, funding, and conflicts of interest are listed at the end of this article.

See also commentary by Gichoya and Trivedi in this issue.

Radiology: Artificial Intelligence 2025; 7(6):e240417 • <https://doi.org/10.1148/ryai.240417> • Content codes: **AI** **BR**

Purpose: To evaluate whether features extracted by Mirai can be aligned with mammographic observations and contribute meaningfully to the prediction of breast cancer risk.

Materials and Methods: This retrospective study examined the correlation of 512 Mirai features with mammographic observations in terms of receptive field and anatomic location. A total of 29 374 screening examinations with mammograms (10 415 female patients; mean age at examination, 60 years $\pm$ 11 [SD]) from the EMory BrEAST imaging Dataset (EMBED) (2013–2020) were used to evaluate feature importance using a feature-centric explainable artificial intelligence pipeline. Risk prediction was evaluated using only calcification features (CalcMirai) or mass features (MassMirai) against Mirai. Performance was assessed in screening and screen-negative (time to cancer, >6 months) populations using the area under the receiver operating characteristic curve (AUC).

Results: Eighteen calcification features and 18 mass features were selected for CalcMirai and MassMirai, respectively. Both CalcMirai and MassMirai had lower performance than Mirai in lesion detection (screening population: Mirai 1-year AUC, 0.81 [95% CI: 0.78, 0.84]; CalcMirai 1-year AUC, 0.76 [95% CI: 0.73, 0.80]; MassMirai 1-year AUC, 0.74 [95% CI: 0.71, 0.78] [$P < .001$]). In risk prediction, there was no evidence of a difference in performance between CalcMirai and Mirai (screen-negative population: Mirai 5-year AUC, 0.66 [95% CI: 0.63, 0.69]; CalcMirai 5-year AUC, 0.66 [95% CI: 0.64, 0.69] [$P = .71$]). However, MassMirai achieved lower performance than Mirai (5-year AUC, 0.57 [95% CI: 0.54, 0.60]; $P < .001$). Radiologist review of calcification features confirmed Mirai's use of benign calcification in risk prediction.

Conclusion: The explainable AI pipeline demonstrated that Mirai implicitly learned to identify mammographic lesion features, particularly calcifications, for lesion detection and risk prediction.

Supplemental material is available for this article.

© The Author(s) 2025. Published by the Radiological Society of North America under a CC BY 4.0 license.

Breast cancer risk prediction has been an important focus of research for decades (1). Traditional risk models (2,3) using epidemiologic data and breast density have been the primary approach to stratify individuals for supplemental screening (4). More recently, however, the need for additional mammographic risk factors beyond breast density has been increasingly recognized (5). Factors such as benign lesion characteristics (6,7) and parenchymal patterns (5) show promise in enhancing risk stratification. Furthermore, the presence of minimal signs and missed cancers at screening mammography (8,9) has prompted discussion about the complementary roles of computer-aided detection (CAD) systems and traditional risk prediction tools (7,10).

Mirai (11,12), a publicly available deep learning (DL) risk model based on mammographic data, has demonstrated superior performance to epidemiologic models that only consider mammographic density. The model's effectiveness has been shown in various clinical contexts: triennial screening (13), high-risk cohorts (14), and when used in conjunction with traditional risk models (15). Notably, Mirai assigns higher risk scores to interval cancers with retrospectively visible lesions (13). This ability to predict malignancy potential has also been indirectly confirmed through selective mirror experiments, in which either the future cancerous or healthy side of the breast was duplicated as model input (14). This highlights Mirai's ability to identify subtle cancer signals that may be overlooked

by radiologists. A derivative simplified model, AsymMirai, that extracts DL features from Mirai and uses bilateral differences in features to predict breast cancer risk demonstrated comparable performance to Mirai (16).

Studies have shown that some extracted features of a DL model correspond to human-interpretable imaging descriptors (17). For example, DL features in a mammographic CAD model align with the Breast Imaging Reporting and Data System (BI-RADS) lexicon (18). However, it remains unclear whether risk prediction models like Mirai share this capability, as mammographic risk factors are not well characterized. As the specific contributions of individual Mirai features to model performance remain largely unexplored, increasing the transparency of Mirai could improve understanding and trust in its risk predictions.

This study aimed to investigate whether a subset of features from the Mirai DL model corresponds to mammographic observations, particularly lesion-related findings, and contributes meaningfully to risk prediction. To achieve this, a feature-centric explainable artificial intelligence (AI) pipeline was developed. First, annotated mammographic observations were linked with Mirai features. Next, partial dependence plot (PDP) (19) analysis was used to examine the influence of selected features on model output. Finally, a feature-masking experiment was conducted to evaluate model performance when restricted to this subset of features.

Abbreviations

AI = artificial intelligence, AUC = area under the receiver operating characteristic curve, BI-RADS = Breast Imaging Reporting and Data System, CAD = computer-aided detection, CSAW-S = Cohort of Screen-Aged Women-S, DL = deep learning, EMBED = EMory BrEast imaging Dataset, KROCC = Kendall rank order correlation coefficient, PDP = partial dependence plot

Summary

A feature-centric explainable artificial intelligence pipeline showed that the Mirai model uses lesion features on mammograms for lesion detection and breast cancer risk prediction. A simplified model using only calcification features achieved comparable risk prediction performance to the original Mirai model in a screen-negative population.

Key Points

- Using a feature-centric explainable artificial intelligence pipeline, features of the deep learning breast cancer risk prediction model Mirai were linked to mammographic observations such as calcifications, masses, and specific breast anatomy. A retrospective study of 29 374 screening examinations (10 415 female patients) found lesion features, especially calcifications, were important elements of Mirai risk prediction within a 5-year time frame.
- A Mirai-derived model using only 18 calcification features (CalcMirai) showed no evidence of a difference in performance compared with Mirai for 5-year risk prediction in a screen-negative population (area under the receiver operating characteristic curve: Mirai, 0.66 [95% CI: 0.63, 0.69] and CalcMirai, 0.66 [95% CI: 0.64, 0.69]; $P = .71$).

Keywords

Breast, Mammography, Screening

Materials and Methods

Datasets

This retrospective study was exempt from institutional review board review due to the use of publicly available datasets. Four publicly available datasets with screening mammograms were used: (a) the Cohort of Screen-Aged Women-S (CSAW-S) dataset (2008 to 2015) (20); (b) the Cyprus mass and (c) the Cyprus calcification datasets (Breast Micro-Calcifications/Masses Dataset with Precisely Annotated Sequential Mammograms, 2009 to 2022) (21–23); and (d) the EMory BrEast imaging Dataset (EMBED; January 2013 to December 2020, publicly available cohort, <https://registry.opendata.aws/emory-breast-imaging-dataset-embed/>) (24). Mirai features were linked to human-labeled mammographic observations from the CSAW-S and both Cyprus datasets, while the EMBED dataset was used to evaluate the importance of selected features to risk prediction. The CSAW-S dataset, originally curated for a segmentation task without cancer information, was used for anatomic mammographic observations, specifically lymph nodes, thick vessels, nipples, and skin. The Cyprus datasets contained sequential mammograms, with the most recent ones indicating malignant masses (retrospective BI-RADS 6) or suspicious calcification findings (retrospective BI-RADS 4 and 5). For simplicity, we selected mammograms with only one annotation for lesions.

EMBED included screening examinations with four mammographic views, craniocaudal and mediolateral oblique views of both breasts, along with biopsy follow-up. Screening

examinations with at least four standard views (captured with Hologic devices, as required by Mirai) were selected. We excluded individuals with breast cancer diagnosis before the first screening examination in the available records and those without at least 1-year follow-up. As the biopsy results are specific to each examination, we could not exclude examinations with prior diagnosis before the curation of the dataset. The subset with follow-up information (risk cohort) was used to calculate risk performance. A subset without follow-up information (background cohort) was used to calculate the mean feature values for the feature masking experiment. To enrich the potential cancer signal intensities, longitudinal examinations of cancer patients (cancer cohort) were selected for the PDP.

Original Mirai Model

The Mirai source code and model weights are publicly available (25). Mirai starts with an encoding process that extracts 512 downscaled feature maps from each mammogram, with each feature map potentially representing mammographic observations learned by the model. The maximum value of each feature map is extracted and defined as the feature value. Pixels contributing to this feature value make up the receptive field. From four mammograms, 4×512 features are aggregated into one image feature vector. Mirai incorporates risk factors from the Tyrer-Cuzick model (2,3) in the final prediction, including *BRCA* mutations, prior benign breast disease, and breast density. If these factors are unavailable, they are estimated from aggregated image features. Finally, an additive hazard layer predicts the 1- to 5-year risk using both image features and risk factors as inputs. As this study focused on mammographic features, we used only the four mammograms from one examination as Mirai input. This is justified as Mirai performance with Tyrer-Cuzick risk factors is not significantly better than without in the original test set (11).

The original Mirai model was trained with screening examinations of first-time cancers and recurrences (11). The training set also included screening-positive examinations that showed obvious lesion signs. During the training process, the image encoder was initially trained to predict risk factors and risks using a single mammogram. The weights were then frozen and utilized to train the entire Mirai model. Consequently, in addition to assessing risk, the Mirai features potentially capture the characteristics of lesions and mammographic findings that correlate with Tyrer-Cuzick risk factors.

Feature-centric Explainable AI Pipeline

A schematic diagram summarizing the feature-centric explainable AI pipeline is shown in Figure 1. This pipeline considered Mirai as containing an encoder followed by a decoder, which includes the image aggregator, risk factor predictor, and additive hazard layer. The pipeline focused on the encoded mammographic features. First, the correlations between Mirai features and mammographic observations were explored, as described in the Linking Mammographic Observations section. Then, we further explored the response of the decoder to the explained features via univariate and multivariate approaches, as described in the Partial Dependence Plot and Mirai Feature Masking sections, respectively.

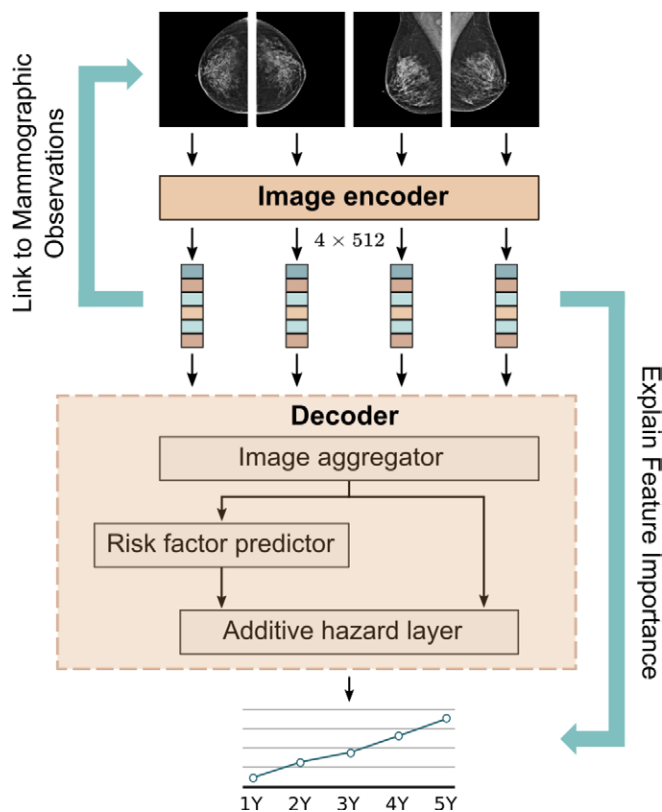


Figure 1: Schematic diagram shows our feature-centric explainable artificial intelligence pipeline. This pipeline first linked Mirai features extracted by the Mirai encoder to mammographic observations. Then, all Mirai components after the encoder were considered as a decoder as a whole. The response of the decoder to the explained Mirai features was explored using a univariate approach (partial dependence plot) and a multivariate approach (Mirai feature masking).

Linking Mammographic Observations

Mammographic observation regions from the CSAW-S and Cyprus datasets were compared with the receptive field centers of 512 features (17), using precision scores (proportion of hits to available annotations) as the metric. Features with the highest precision by point estimate in anatomic mammographic observations (lymph nodes, thick vessels, nipples, and skin) were selected. Recognizing the clinical relevance of lesion features, we decided not to limit our selection to just one feature for calcification and mass. Instead, we selected the same number of best lesion features based on their precision, ensuring consistent degrees of freedom for decoder-response comparison. Initially, we identified the minimum set of features required to achieve precision scores greater than or equal to 0.5 for calcification and mass individually, implying that the selected feature located the mammographic observation in at least half of the cases. We then used the smaller number of features from these two sets, reflecting our focus on the minimal information needed to maintain the original performance level.

Partial Dependence Plot

The PDP (19) examined the relationship between each feature and the resulting risk prediction (1- to 5-year risk) while holding all other features constant. For a feature of interest, the feature value was varied from the fifth to the 95th percentile of the available values of this feature across the data

cohort. The absolute risk change relative to the risk at the fifth percentile feature value was calculated for each percentile and averaged across the population. Feature importance was derived from the PDP scores, using the risk at the 95th percentile feature value (Appendix S1). As each view was encoded with the same encoder, PDPs for 512 features were analyzed separately for all four views. The target population was the EMBED cancer cohort.

Mirai Feature Masking

To assess decoder performance limited to these newly derived lesion features, two constrained Mirai variants were created: CalcMirai, using only calcification features, and MassMirai, using only mass features. This was achieved using the original Mirai decoder but with masking the other features with their corresponding mean values from the EMBED background cohort. Prediction performance was evaluated with the EMBED risk cohort and compared with the original Mirai model. Under this setting, Mirai variants only had population-level information from the EMBED background cohort for nonlesion features, while the original Mirai model had individual-level information for these features in the final test set.

The impact of different numbers of selected features was assessed by a sensitivity analysis (Appendix S2). Subsequently, subgroup analyses were conducted based on age, density, and lesion types at diagnosis (Appendix S3). Risk performance was also evaluated on examinations with a time to diagnosis longer than 6 months, serving as a surrogate for screen-negative examinations (11,12). Our analyses focused on two regimens: the lesion detection regimen (screening population, 1-year risk) and the risk prediction regimen (screen-negative population, 5-year risk).

Radiologist Review of Calcification Features

Based on preliminary analyses, we conducted a radiologist review of the receptive fields of selected features in CalcMirai. Screen-negative examinations with future cancer diagnoses within 5 years and a CalcMirai risk score in the top 10% were selected from the EMBED risk cohort. An expert breast radiologist (more than 15 years of experience) evaluated calcifications identified by Mirai and classified them into subgroups of “typically benign” and other calcifications based on their distribution and morphology in the BI-RADS categories (Appendix S4).

Statistical Analysis

Encoder features were ordered using PDP scores applied on the different views, with ranks compared using the Kendall rank order correlation coefficient (KROCC). The bootstrap method, sampling 1000 times, was used to estimate the CIs of correlation coefficients, as described in the PDP section, and precision, as described in the Linking Mammographic Observations section. Model performance was evaluated using the area under the receiver operating characteristic curve (AUC). The CIs and differences between paired AUC values were calculated using the DeLong method (26). All tests were two-sided, and a *P* value of < .05 was considered significant. Statistical analyses were performed using Python 3.9.16 and SciPy 1.9.1. All code is available at <https://gitlab.kuleuven.be/medphysqa/deploy/xai-mirai-features>.

Table 1: Demographics of Cyprus Datasets

Parameter	Calcification	Mass
Age (y)		
<50	2/46 (4)	2/37 (5)
50–60	18/46 (39)	7/37 (19)
60–70	23/46 (50)	21/37 (57)
70–80	2/46 (4)	6/37 (16)
80–90	1/46 (2)	1/37 (3)
Density		
BI-RADS a	4/46 (9)	4/37 (11)
BI-RADS b	23/46 (50)	16/37 (43)
BI-RADS c	16/46 (35)	16/37 (43)
BI-RADS d	3/46 (7)	1/37 (3)
BI-RADS assessment		
BI-RADS 4A	25/46 (54)	0
BI-RADS 4B	15/46 (33)	0
BI-RADS 4C	5/46 (11)	0
BI-RADS 5	1/46 (2)	0
BI-RADS 6	0	37/37 (100)
Biopsy finding		
Benign	20/46 (43)	0
Malignant	8/46 (17)	37/37 (100)
Malignant (DCIS)	4/46 (9)	NA
Unknown	18	0
No. of individuals	46	37

Note.—Data are either numbers or numerators and denominators, with percentages in parentheses. *Cyprus datasets* refer to Breast Micro-Calcifications and Masses Dataset with Precisely Annotated Sequential Mammograms. BI-RADS = Breast Imaging Reporting and Data System, DCIS = ductal carcinoma in situ, NA = not available.

Results

Patient Demographics

A total of 336 mammograms from 172 female patients were included from the CSAW-S dataset, with no demographic information available. From the Cyprus mass and calcification datasets, 62 mammograms from 37 female patients and 90 mammograms from 46 female patients were selected, respectively, with an average age of 61 years $\pm$ 7 (SD) and 63 years $\pm$ 7. All mass annotations were biopsy-proven malignant, while only eight calcification annotations were malignant, with four being ductal carcinoma in situ. Other demographic details are summarized in Table 1.

The EMBED risk cohort comprised 29,374 examinations from 10,415 female patients with follow-up data (mean age, 60 years $\pm$ 11; Fig 2). A total of 7,349 examinations from 7,234 female patients (mean age, 55 years $\pm$ 12) in the EMBED background cohort were used to determine the mean feature values. The EMBED cancer cohort was a subset of the risk cohort, containing 644 examinations from 263 female patients (mean age, 62 years $\pm$ 11). Table 2 summarizes the demographic and lesion details of the EMBED cohorts.

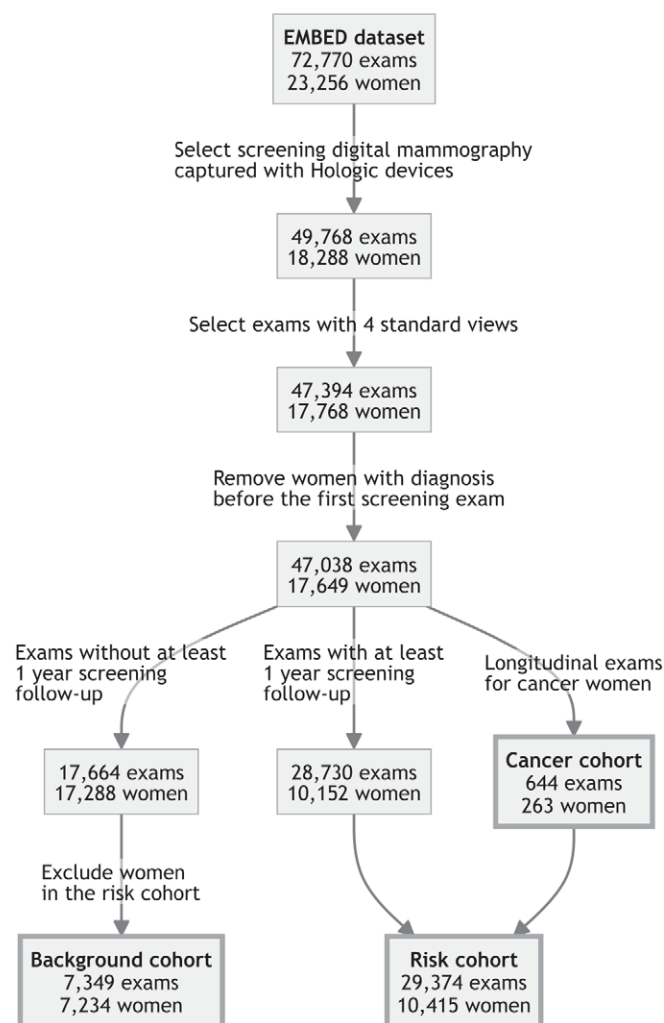


Figure 2: Flowchart shows the data exclusion for EMory BrEast imaging Dataset (EMBED). The risk, cancer, and background cohorts were utilized in this study. The removal of female patients with a diagnosis before the first screening examination during the EMBED curation timeframe was to avoid confusion in defining positive and negative risk cases. Due to the dataset label limitation, we could not exclude examinations with a prior history of breast cancer.

Linking Mammographic Observations Results

Figure 3A illustrates the encoder feature extraction process. The receptive fields of Mirai features that were associated with the highest precision to mammographic observations are shown in Figure 3B. The precision score between feature 145 and suspicious calcification clusters was 0.81 (73 of 90 [95% CI: 0.73, 0.89]), while that for feature 371 and malignant mass regions was 0.79 (49 of 62 [95% CI: 0.68, 0.89]). These were the mammographic observations with precision scores higher than 0.7. However, other anatomic features with low precision were still linked to structures that share a similar visual appearance.

For lesions, features with precision scores greater than or equal to 0.5 were selected, resulting in 18 calcification features (precision from 0.54 to 0.81) and 42 mass features (precision from 0.50 to 0.79). To finalize the selection, the minimum number of lesion features meeting the precision threshold was chosen, resulting in 18 calcification and 18 mass features (lowest precision of 0.66).

Table 2: Demographic and Lesion Characteristics of Selected Cohorts in EMBED

Parameter	Risk Cohort		Background Cohort
	Individual-based	Examination-based	Examination-based
Age (y)			
<50	NA	6284/29 374 (21)	2972 (40)
50–60	NA	8031/29 374 (27)	2126 (29)
60–70	NA	8850/29 374 (30)	1403 (19)
70–80	NA	5107/29 374 (17)	637 (9)
80–90	NA	1025/29 374 (3)	186 (3)
Unknown	NA	77	25
Density			
BI-RADS a	NA	3400/29 374 (12)	741 (10)
BI-RADS b	NA	12 285/29 374 (42)	2988 (41)
BI-RADS c	NA	12 069/29 374 (41)	3209 (44)
BI-RADS d	NA	1537/29 374 (5)	394 (5)
Unknown	NA	83	17
Lesion*			
Calc	117/263 (44)	272/644 (42)	NA
Mass	84/263 (32)	185/644 (29)	NA
Asym	96/263 (37)	238/644 (37)	NA
AD	43/263 (16)	110/644 (17)	NA
Calc and mass	16/263 (6)	24/644 (4)	NA
Calc and asym	22/263 (8)	49/644 (8)	NA
Calc and AD	10/263 (4)	14/644 (2)	NA
Mass and asym	32/263 (12)	65/644 (10)	NA
Mass and AD	8/263 (3)	18/644 (3)	NA
Asym and AD	10/263 (4)	27/644 (4)	NA
Cancer side			
Left	118/263 (45)	298/644 (46)	NA
Right	134/263 (51)	328/644 (51)	NA
Bilateral	11/263 (4)	18/644 (3)	NA
Diagnosis	263 female patients	644 examinations	NA
Summation	10 415 female patients	29 374 examinations	7349 examinations 7234 female patients

Note.—Unless otherwise noted, data are either numbers or numerators and denominators, with percentages in parentheses. AD = architectural distortion, asym = asymmetry, BI-RADS = Breast Imaging Reporting and Data System, calc = Calcification, EMBED = EMory BrEast imaging Dataset, NA = not applicable.

* An examination might discover multiple lesions at the time of diagnosis. Each lesion group is not exclusive. For example, the calcification group could include examinations with both calcification and mass.

PDP Scores

The PDP scores were extracted from four mammographic views targeting 1- to 5-year risk prediction. The scores showed a high correlation across different projections (eg, craniocaudal vs mediolateral oblique), with KROCC values between 0.96 (95% CI: 0.95, 0.97) and 0.97 (95% CI: 0.96, 0.98) ($P < .001$) and an even stronger correlation between identical projections (eg, right and left craniocaudal), with KROCC values between 0.987 (95% CI: 0.98, 0.99) and 0.995 (95% CI: 0.994, 0.997) ($P < .001$). Within one projection view, but for different prediction time points (eg, 1- and 5-year risk), the KROCC values had correlations of between 0.97 (95% CI: 0.95, 0.98) and 0.997 (95% CI:

0.995, 0.999) ($P < .001$). The PDPs of the right mediolateral oblique view targeting 1-year risk are selected for demonstration in Figure 4.

Among all mammographic observations in Figure 4A, the calcification feature had the largest impact on prediction compared with others. It is striking that the mass lesion feature with the highest precision had a similar impact compared with other anatomic structures. This finding is clearly visualized in Figure 4B and 4C, where 18 selected calcification and mass features are plotted, demonstrating the importance of calcification features not only across identified mammographic observations but also across all features.

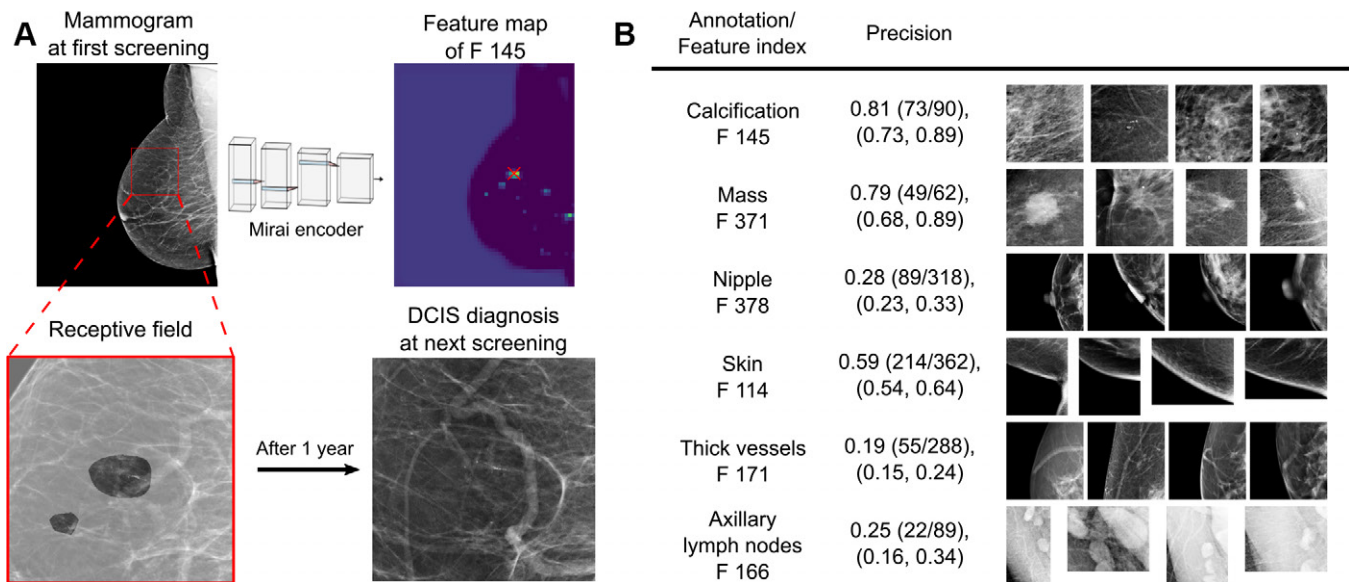


Figure 3: Association of mammographic observations with features using receptive fields extracted from the Mirai encoder. **(A)** Figure shows an example for calcifications with feature (F) 145. The Mirai encoder first extracts a downsized feature map from a mammogram. The maximum value (marked by a red cross) is taken as the final feature value. The thresholded feature map is overlaid on the receptive field as a visual guideline. The selected receptive field was confirmed as ductal carcinoma in situ (DCIS) after 1 year. The image source is the Cyprus calcification (Breast Micro-Calcifications Dataset with Precisely Annotated Sequential Mammograms) dataset. The mammogram of a 68-year-old female patient was acquired with a right mediolateral oblique view. **(B)** Figure shows feature indexes and receptive fields associated with mammographic observations. Precision scores, along with the number of associations and available mammographic observation regions and 95% CIs, are reported.

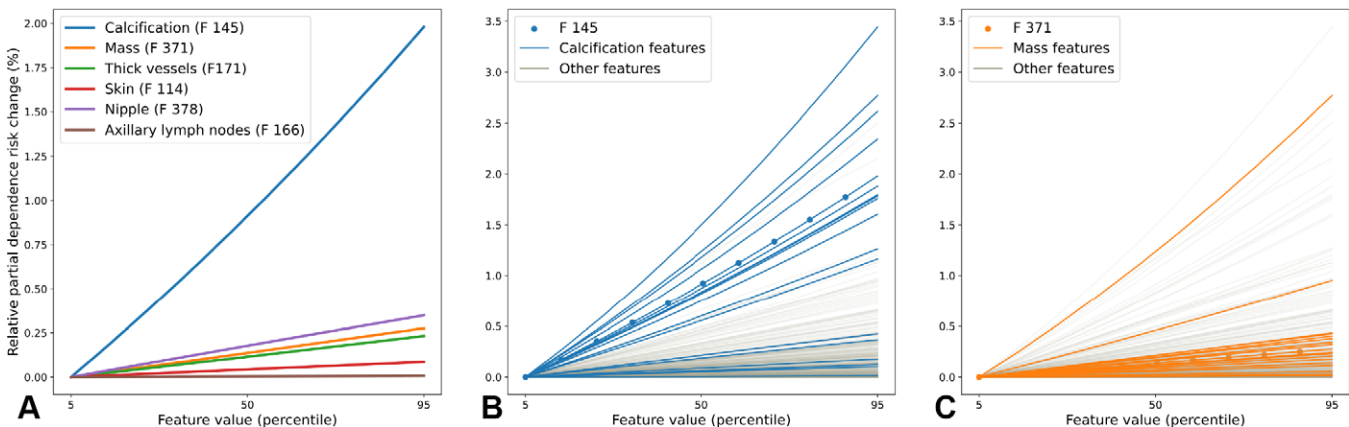


Figure 4: Partial dependence plots show absolute risk change relative to the risk at the fifth percentile, plotted as a function of percentile, calculated on the right mediolateral oblique view, targeted at 1-year risk. **(A)** Plot shows identified features (Fs) associated with mammographic observations with the highest precision. **(B)** Plot shows selected calcification features. **(C)** Plot shows selected mass features.

Mirai Feature Masking Results

The Mirai decoder was tested separately for 18 calcification (CalcMirai) and 18 mass features (MassMirai) for each view. Model performance, evaluated with the EMBED risk cohort, is presented in Table 3. In the lesion detection regimen (screening population, 1-year risk), the original Mirai model exhibited the highest AUC values among the three models, as expected. Interestingly, although both CalcMirai and MassMirai had lower performance than Mirai (Mirai AUC, 0.81 [95% CI: 0.78, 0.84]; CalcMirai AUC, 0.76 [95% CI: 0.73, 0.80]; MassMirai AUC, 0.74 [95% CI: 0.71, 0.78]; $P < .001$), both had 95% CIs overlapping with Mirai using only 3.5% (18 of 512) of the features. In the risk-prediction regimen (screen-negative population, 5-year risk), there was no evidence of a difference in performance between CalcMirai and Mirai (AUC: Mirai AUC, 0.66 [95% CI: 0.63, 0.69] vs CalcMirai AUC, 0.66 [95% CI:

0.64, 0.69]; $P = .71$). However, MassMirai underperformed compared with both Mirai and CalcMirai (AUC, 0.57 [95% CI: 0.54, 0.60]; P value to Mirai $< .001$).

The sensitivity analysis on the number of included features, as shown in Figure 5, further supports the observation that both mass and calcification features are important for risk prediction, even when considering the inclusion of 42 mass features that meet the precision threshold. Subgroup analyses (Appendix S3, Table S1) across lesion types at diagnosis confirmed that CalcMirai and MassMirai were not only effective within their respective lesion groups but also demonstrated applicability to other lesion types.

Radiologist Review of Calcification Features

The results of the radiologist review of receptive fields, identified by calcification features, are summarized in Table 4, and repre-

Table 3: Comparison of Model Performance in Mirai Feature-masking Experiment

Parameter	1-Year AUC	2-Year AUC	3-Year AUC	4-Year AUC	5-Year AUC
Screening population					
<i>n</i> +/ <i>n</i> -	232/29 142	354/20 719	453/14 197	526/90 12	588/51 14
Mirai	0.81 (0.78, 0.84)	0.77 (0.74, 0.79)	0.75 (0.73, 0.77)	0.74 (0.71, 0.76)	0.72 (0.70, 0.75)
CalcMirai	0.76 (0.73, 0.80)	0.73 (0.70, 0.76)	0.71 (0.69, 0.74)	0.71 (0.68, 0.73)	0.70 (0.68, 0.73)
<i>P</i> value	<.001	<.001	<.001	<.001	.009
MassMirai	0.74 (0.71, 0.78)	0.69 (0.66, 0.72)	0.66 (0.63, 0.69)	0.65 (0.62, 0.67)	0.63 (0.61, 0.66)
<i>P</i> value	<.001	<.001	<.001	<.001	<.001
Screen-negative population					
<i>n</i> +/ <i>n</i> -	17/29 142	139/20 719	238/14 197	311/90 12	373/51 14
Mirai	0.70 (0.57, 0.82)	0.68 (0.64, 0.72)	0.68 (0.65, 0.71)	0.67 (0.64, 0.70)	0.66 (0.63, 0.69)
CalcMirai	0.68 (0.56, 0.80)	0.66 (0.62, 0.71)	0.66 (0.63, 0.70)	0.67 (0.64, 0.70)	0.66 (0.64, 0.69)
<i>P</i> value	.59	.31	.16	.46	.71
MassMirai	0.65 (0.52, 0.77)	0.60 (0.55, 0.64)	0.58 (0.54, 0.62)	0.58 (0.55, 0.61)	0.57 (0.54, 0.60)
<i>P</i> value	.32	<.001	<.001	<.001	<.001

Note.—Unless otherwise noted, data are numerators and denominators or areas under the receiver operating characteristic curve values (AUCs), with 95% CIs in parentheses. The risk cohort from EMory BrEast imaging Dataset (EMBED) was used to evaluate the performance of the Mirai model with calcification-only Mirai features (CalcMirai) and mass-only Mirai features (MassMirai). To avoid obvious lesion signs, the screen-negative population was also investigated. The number of positive (accumulated up to the indicated year) and negative examinations in each group is denoted as “*n*+/*n*-.” Each *P* value for CalcMirai and MassMirai corresponds to the significance of the performance difference when compared with the original Mirai model.

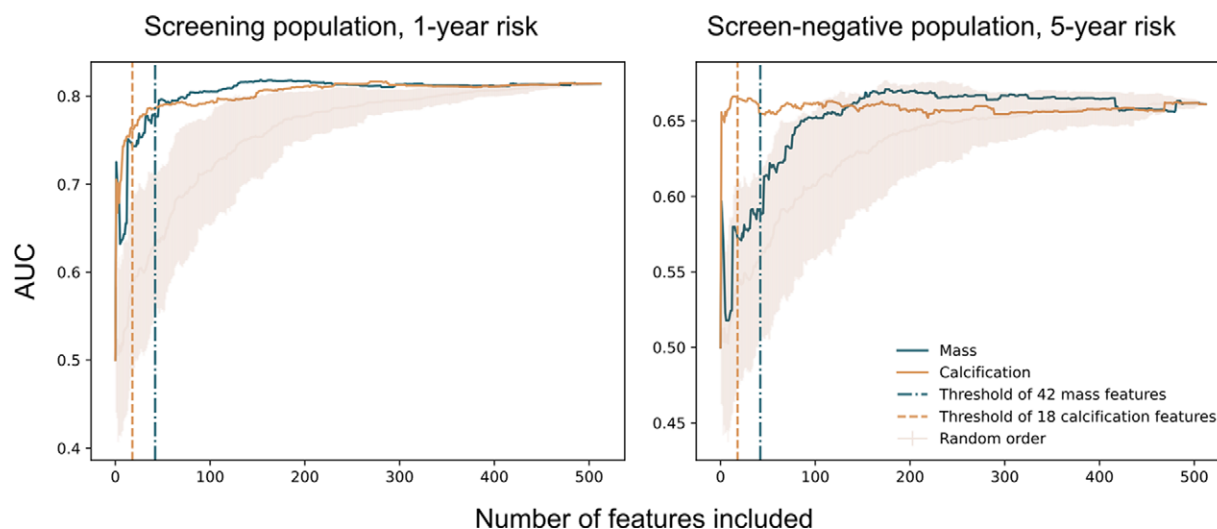


Figure 5: Plots show sensitivity analysis on the number of features included in Mirai lesion variants CalcMirai and MassMirai for the lesion-detection regimen (screening population, 1-year risk) and the risk-prediction regimen (screen-negative population, 5-year risk). Model performance was evaluated using the area under the receiver operating characteristic curve (AUC). In addition to features ordered by lesion precision, 40 random orderings were generated as controls. Features were incrementally added to the Mirai variants, from fully masked (0 features used) to fully included (equivalent to Mirai). Both thresholds for mass and calcification features with precision scores greater than or equal to 0.5 were indicated. Finally, 18 features were selected for both Mirai variants.

Table 4: Radiologist Review on Receptive Fields of 18 Mirai Calcification Features

Calcification Type	No. of Receptive Fields
Typically benign	511*
Round and punctate	230
Large rod-like/secretory	83
Default (not specified)	63
Coarse/popcorn-like	62
Vascular	29
Fat necrosis	24
Skin	9
Milk of calcium	7
Postsurgical	4
No calcification	311
Other calcification	128
Summation	950

Note.—*No calcification* includes surgical clips and areas outside the breast. *Other calcification* indicates that the calcification needs to be described by its distribution and morphology. *Typically benign* calcifications are further categorized into subtypes.

* Indicates the total number of all typically benign subtypes.

sentative cases are shown in Figure 6. In total, 14 screen-negative examinations were selected: 10 examinations diagnosed within 2 years (short term) and four examinations diagnosed between 4 and 5 years (long term). A total of 950 receptive fields of calcification features were evaluated by the radiologist. In the short-term group, the reading results suggested that three examinations had BI-RADS class 4 microcalcifications, three examinations had calcifications within suspicious masses, and one examination had a suspicious mass alone. The other detected calcifications were all benign, as expected. Among all the benign calcifications, punctate calcifications were most frequently captured by the calcification features, which could be suspicious if accompanied by a certain distribution. Interestingly, in addition to calcifications, receptive fields associated with calcification features also contained surgical clips.

Discussion

To enhance transparency of the Mirai DL model, we developed a feature-centric explainable AI pipeline to investigate the extent to which encoder features in Mirai correspond to annotated mammographic observations. Calcification, mass, and anatomic features, such as nipple, could be linked to specific Mirai features. Recognizing the clinical relevance of lesions for risk prediction, we built two Mirai variants, CalcMirai and MassMirai, each utilizing only 18 features (3.5% of all features) specific to calcification and mass, respectively. Mirai's performance was tested when limited to these selected features. Our results suggest that Mirai implicitly performs two tasks, lesion detection and risk prediction, using lesion features. In the lesion detection regimen (screening population, 1-year risk), both CalcMirai and MassMirai underperformed compared with Mirai. However, in the risk prediction regimen (screen-negative population, 5-year risk), CalcMirai performed similarly to Mirai

and outperformed MassMirai. The results from the radiologist review suggest that the long-term risk prediction of Mirai might depend on benign calcifications.

As the original Mirai training set included screen-positive examinations (11), the model might have learned features that could be shared for both lesion detection and risk prediction. This is evident from Mirai features aligning with lesion and anatomic mammographic observations, including also diagnostic indicators such as skin thickening, nipple retraction, and lymphadenopathy (27). In particular, the importance of lesion features for Mirai was more pronounced than that of other anatomic features. The success of CalcMirai and MassMirai might be attributed to the high prevalence of calcifications and masses in cancer examinations, as well as the utility of these features in both lesion detection and risk prediction for other lesion types. For example, MassMirai could detect architectural distortion, implying that mass features might also capture the parenchymal patterns surrounding masses.

Our study clearly shows that lesion signal contributes to the performance of a risk predictor. Mirai's malignant lesion detection component, which has been shown to outperform some CAD systems (10), allows Mirai to find missed lesions at screen-negative examinations. The role of benign lesions remains unclear, as radiologists primarily focus on lesion detection. The observed correlation between benign calcifications and risk could be explained by the fact that benign breast diseases are known risk factors (28,29). Our finding of Mirai leveraging benign lesions as risk factors is consistent with another risk model (6,7) that directly includes lesion scores.

We acknowledge that there may be confounding factors within our selected features and that important mammographic observations may not have been included. Four of the 18 features were common to both mass and calcification features. This may result from masses often being accompanied by microcalcifications. There is no evidence to reject the existence of other strong risk factors, such as parenchymal patterns (5,30), which have already been researched before the advent of DL models. In addition, the long-term risk prediction component of calcification features might be influenced by parenchymal patterns around the calcifications. However, our results suggest that identifying risk-related parenchymal patterns through data-driven models requires careful data curation to exclude subtle calcification signs, as calcifications would be a strong confounding risk factor in such experiments.

A previous study, introducing AsymMirai, showed that Mirai predicted uniformly low risk scores after mirroring the unilateral breast (16) and further concluded that local bilateral dissimilarity was key in the Mirai prediction. However, a similar study (14) applied selective mirroring based on the future cancerous side. Using only the cancerous breast achieved the same performance compared with no mirroring, despite the model giving low risk scores similar to the observation with AsymMirai. This suggests asymmetry might not be central in Mirai's risk estimation. As the features extracted in our study are the same as those extracted in AsymMirai, another explanation for the good performance of AsymMirai is the presence of lesion signs. Considering the low incidence of bilateral breast cancer, AsymMirai's bilateral difference may actually capture the relative malignant tendency between the breasts.

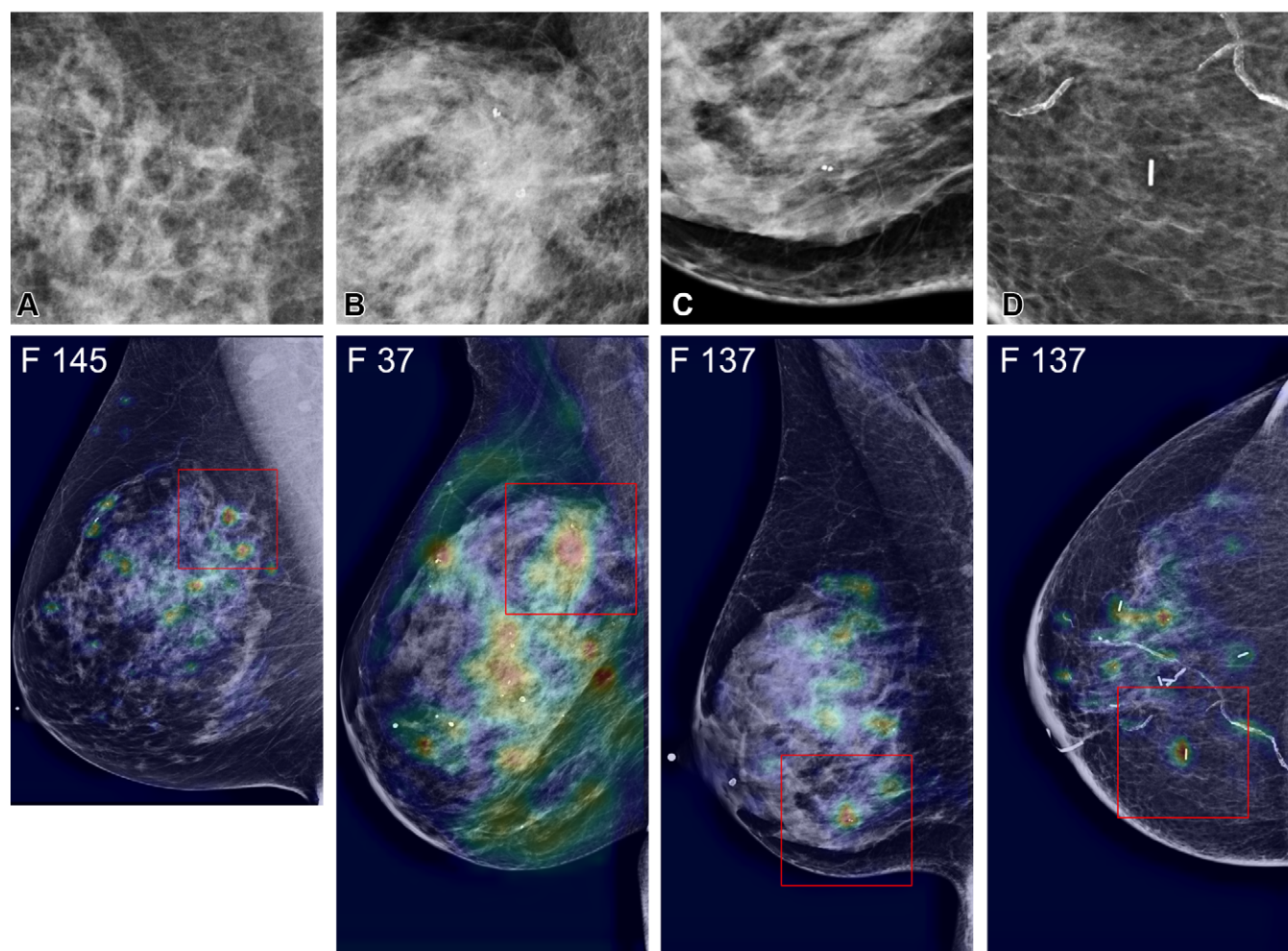


Figure 6: Mammograms show representative cases from the radiologist review of calcification features. Four receptive fields from negative screening mammograms of female patients, each containing different objects, are selected: Images show (A) suspicious calcification (right mediolateral oblique view; patient age, 68 years), (B) benign calcification within a suspicious mass (right mediolateral oblique view; patient age, 69 years), (C) benign calcification (left mediolateral oblique view, flipped to the right to meet Mirai requirements; patient age, 68 years), and (D) surgical clip (right craniocaudal view; patient age, 76 years). The first row of images shows the identified receptive fields. The second row shows the overlap of feature maps on the whole mammogram, with white text denoting the corresponding Mirai feature (F) and red squares indicating the location of the receptive fields. The images' source is the EMory BrEast imaging Dataset (EMBED).

Our study had limitations. Although calcification and mass annotations were curated for the same purpose, the inclusion criteria were different, as there were no suspicious masses with benign biopsy. This could lead to bias in the selected features. Additionally, the lack of annotations for other lesion types limited the exploration of all lesion features. This is evident from the fact that neither CalcMirai nor MassMirai could achieve the performance of Mirai for all lesion subgroups. Finally, breast density, a well-established risk factor, was not included. The receptive field cannot cover the entire breast, and therefore, density information might not be fully captured by individual features.

In conclusion, the explainable AI framework presented in this study demonstrates that the Mirai risk model implicitly learned to identify mammographic observations not explicitly defined during training. It also provides direct evidence that lesion features contribute to both lesion detection and risk prediction tasks. Notably, the simplified CalcMirai model, which used only 3.5% of the available features, achieved performance comparable to that of the full Mirai model. Further studies are required to explore the added value of short-term (screening interval) risk prediction models over CAD systems.

Author affiliations:

¹ Department of Imaging and Pathology, KU Leuven, Herestraat 49, Box 7003, 3000 Leuven, Belgium

² Faculty of Mathematics and Physics, University of Ljubljana, Ljubljana, Slovenia

³ Department of Radiology, University Hospital Leuven, Leuven, Belgium

⁴ Jožef Stefan Institute, Ljubljana, Slovenia

⁵ Department of Medical Physics, University of Wisconsin-Madison, Madison, WI

Received July 8, 2024; revision requested September 15; revision received July 22, 2025; accepted August 19.

Address correspondence to: H.B. (email: hilde.bosmans@uzleuven.be).

Supplemental material: Supplemental material is available at *Radiology: Artificial Intelligence* online.

Funding: This work was supported by the Research Foundation-Flanders (research core funding FWO G0A7121N) and the Slovenian Research Agency (research core funding P1-0389).

Acknowledgments: We extend our thanks to Valerie Celis, MD, for her expertise in the radiologist review of calcification features, and to Frederik De Keyzer, MSc, for reviewing the manuscript. Our research benefited significantly from the use of the open source Mirai model and the public dataset collection, including the Cyprus mass and calcification datasets (Breast Micro-Calcifications/Masses Dataset with Precisely Annotated Sequential Mammograms), the CSAW-S dataset, and EMBED.

Author contributions: Guarantors of integrity of entire study, **Y.K.W., H.B.**; study concepts/study design or data acquisition or data analysis/interpretation, all authors; manuscript drafting or manuscript revision for important intellectual content, all authors; approval of final version of submitted manuscript, all authors; agrees to ensure any questions related to the work are appropriately resolved, all authors; literature research, **Y.K.W., N.M., A.S., H.B.**; clinical studies, **Y.K.W., H.B.**; experimental studies, **Y.K.W., L.C., H.B.**; statistical analysis, **Y.K.W., R.J., H.B.**; and manuscript editing, all authors

Disclosures of conflicts of interest: **Y.K.W.** No relevant relationships. **Z.K.** No relevant relationships. **T.W.** No relevant relationships. **L.C.** No relevant relationships. **N.M.** No relevant relationships. **A.S.** Payment from the National Review Commission for Reviewing Public Procurement Award Procedures for a review of detection techniques in mammography. **R.J.** No relevant relationships. **H.B.** Employed by the KU Leuven and the University Hospital of Leuven (in the frame of this employment, author took the lead in the current research project); membership on the board of directors of the CVKO (center for cancer screening) in Flanders, Belgium (royalties go to the University Hospital Leuven).

References

- Shieh Y, Eklund M, Madlensky L, et al; Athena Breast Health Network Investigators. Breast cancer screening in the precision medicine era: Risk-based screening in a population-based trial. *J Natl Cancer Inst* 2017;109(5).
- Tyrer J, Duffy SW, Cuzick J. A breast cancer prediction model incorporating familial and personal risk factors. *Stat Med* 2004;23(7):1111–1130.
- Tyrer J, Duffy SW, Cuzick J. IBIS breast cancer risk evaluation tool. <https://ems-trials.org/riskevaluator/>. Updated September 17, 2017.
- Monticciolo DL, Newell MS, Moy L, Lee CS, Destounis SV. Breast cancer screening for women at higher-than-average risk: Updated recommendations from the ACR. *J Am Coll Radiol* 2023;20(9):902–914.
- Gastouni A, Conant EF, Kontos D. Beyond breast density: A review on the advancing role of parenchymal texture analysis in breast cancer risk assessment. *Breast Cancer Res* 2016;18(1):91.
- Eriksson M, Czene K, Pawitan Y, Leifland K, Darabi H, Hall P. A clinical model for identifying the short-term risk of breast cancer. *Breast Cancer Res* 2017;19(1):29.
- Azam S, Eriksson M, Sjölander A, et al. Mammographic microcalcifications and risk of breast cancer. *Br J Cancer* 2021;125(5):759–765.
- Hovda T, Tsuruda K, Hoff SR, Sahlberg KK, Hofvind S. Radiological review of prior screening mammograms of screen-detected breast cancer. *Eur Radiol* 2021;31(4):2568–2579.
- Hovda T, Hoff SR, Larsen M, Romundstad L, Sahlberg KK, Hofvind S. True and missed interval cancer in organized mammographic screening: A retrospective review study of diagnostic and prior screening mammograms. *Academic Radiology* 2022;29:S180–S191.
- Santeramo R, Damiani C, Wei J, Montana G, Brentnall AR. Are better AI algorithms for breast cancer detection also better at predicting risk? A paired case–control study. *Breast Cancer Res* 2024;26(1):25.
- Yala A, Mikhael PG, Strand F, et al. Toward robust mammography-based models for breast cancer risk. *Sci Transl Med* 2021;13(578):eaba4373.
- Yala A, Mikhael PG, Strand F, et al. Multi-institutional validation of a mammography-based breast cancer risk model. *J Clin Oncol* 2022;40(16):1732–1740.
- Damiani C, Kalliatakis G, Sreenivas M, et al. Evaluation of an AI model to assess future breast cancer risk. *Radiology* 2023;307(5):e222679.
- Omoleye OJ, Woodard AE, Howard FM, et al. External evaluation of a mammography-based deep learning model for predicting breast cancer in an ethnically diverse population. *Radiol Artif Intell* 2023;5(6):e220299.
- Arasu VA, Habel LA, Achacoso NS, et al. Comparison of mammography AI algorithms with a clinical risk model for 5-year breast cancer risk prediction: An observational study. *Radiology* 2023;307(5):e222733.
- Donnelly J, Moffett L, Barnett AJ, et al. AsymMirai: Interpretable mammography-based deep learning model for 1–5-year breast cancer risk prediction. *Radiology* 2024;310(3):e232780.
- Bau D, Zhou B, Khosla A, Oliva A, Torralba A. Network dissection: Quantifying interpretability of deep visual representations. *IEEE conference on computer vision and pattern recognition (CVPR)*. Honolulu, HI: IEEE 2017; 319–3327.
- Wu J, Zhou B, Peck D, et al. DeepMiner: Discovering interpretable representations for mammogram classification and explanation. *Harvard Data Science Review* 2021;3(4).
- Friedman JH. Greedy function approximation: A gradient boosting machine. *Ann Statist* 2001;29(5).
- Matsoukas C, Hernandez ABI, Liu Y, et al. Adding Seemingly Uninformative Labels Helps in Low Data Regimes. *arXiv* 2020. Preprint posted online June 20, 2020; <http://arxiv.org/abs/2008.00807>.
- Loizidou K, Skouroumouni G, Pitris C, Nikolaou C. Digital subtraction of temporally sequential mammograms for improved detection and classification of microcalcifications. *Eur Radiol Exp* 2021;5(1):40.
- Loizidou K, Skouroumouni G, Savvidou G, et al. Breast Masses Dataset with Precisely Annotated Sequential Mammograms v2. Zenodo. <https://zenodo.org/records/11446259>. Updated June 4, 2024.
- Loizidou K, Skouroumouni G, Nikolaou C, Pitris C. Breast Micro-Calcifications Dataset with Precisely Annotated Sequential Mammograms. Zenodo. <https://zenodo.org/records/7969411>. Published June 30, 2021.
- Jeong JJ, Vey BL, Bhimireddy A, et al. The EMory BrEAST imaging dataset (EMBED): A racially diverse, granular dataset of 3.4 million screening and diagnostic mammographic images. *Radiol Artif Intell* 2023;5(1):e220047.
- Yala A. Mirai. GitHub repository. <https://github.com/yala/Mirai>.
- DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics* 1988;44(3):837–845.
- Sickles EA, D'Orsi CJ, Bassett LW, et al. ACR BI-RADS Mammography. Reston, VA: American College of Radiology; 2013.
- Dyrstad SW, Yan Y, Fowler AM, Colditz GA. Breast cancer risk associated with benign breast disease: Systematic review and meta-analysis. *Breast Cancer Res Treat* 2015;149(3):569–575.
- Tse GM, Tan PH, Pang ALM, Tang APY, Cheung HS. Calcification in breast lesions: Pathologists' perspective. *J Clin Pathol* 2008;61(2):145–151.
- Acciavatti RJ, Lee SH, Reig B, et al. Beyond breast density: Risk measures for breast cancer in multiple imaging modalities. *Radiology* 2023;306(3):e222575.